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Increasing the Resolution of the Inversion by Combining Deep Resistivity with Ultra-Deep Resistivity

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Abstract

This paper explores the enhancement of 1D inversion resolution by integrating Deep Azimuthal Resistivity (DAR) and Ultra-Deep Azimuthal Resistivity (UDAR) data. The primary aim is to improve the resolution of resistivity models, focusing on the identification of thin layers and increased depth of detection (DOD). The enhanced resolution in 1D inversion supports more accurate geological interpretation, particularly in reservoirs with complex layering and subtle petrophysical variations.

The workflow employs a hybrid inversion approach that combines DAR and UDAR measurements. This method leverages parameter domain scanning and forward modeling to refine inversion results. By increasing the depth of detection through UDAR, and using DAR to capture finer resistivity contrasts, the inversion process enhances the resolution of the model. Misfit analysis and azimuthal signal variations across frequencies help identify and address multi-dimensional influences, enabling more accurate 1D resistivity profiles. The resulting models are generated with ultra-high-definition (UHD) resolution, allowing for a more precise interpretation of subsurface features, particularly thin layers and boundary delineations.

The use of enhanced resolution in 1D inversion helps address challenges in complex reservoirs with fine-scale variations, allowing for better detection of thin layers and more precise boundary delineation. The methodology offers a significant improvement in subsurface mapping and enhances the overall reliability of reservoir evaluations.

Introduction

Accurate subsurface characterization is crucial for successful hydrocarbon exploration and production, particularly in complex geological environments. Among various subsurface evaluation techniques, resistivity logging has long played a critical role in formation evaluation, reservoir characterization, and geosteering. Resistivity measurements provide essential insights into fluid content, rock types, and geological structures, thereby aiding in accurate reservoir delineation and effective resource exploitation.

Over the decades, resistivity logging technology has significantly evolved, progressing from simple induction and laterolog tools to advanced multi-frequency, azimuthal, and deep-reading resistivity systems.

Standard resistivity tools traditionally measure formation properties within a limited radial depth, typically within inches to a few feet around the borehole. While these tools are effective in relatively homogeneous and predictable geological settings, their limited depth of investigation often restricts their usefulness in structurally complex and heterogeneous reservoirs.

In response to the growing demands for improved subsurface resolution, particularly in intricate geological formations, advancements in deep and ultra-deep resistivity logging technologies have emerged. These advanced tools offer significantly greater depths of investigation, ranging from several meters to tens of meters around the borehole, enabling earlier and more precise detection of formation boundaries, fluid contacts, and geological anomalies. This capability provides operators with invaluable early insights, allowing proactive geosteering decisions and significantly reducing drilling risks and uncertainties.

key innovation driving these advanced resistivity logging tools is the sophisticated integration of multiple antenna configurations, frequencies, and orientations, coupled with powerful computational inversion techniques. Among these innovations, Deep Azimuthal Resistivity and Ultra-Deep Azimuthal Resistivity technologies have become instrumental due to their extended depth of investigation and directional sensitivity. These technologies generate large datasets characterized by high spatial resolution and azimuthal sensitivity, which are critical for accurately resolving thin beds, subtle geological structures, and complex stratigraphic features.

Moreover, the combination of deep and ultra-deep resistivity data, when integrated into advanced inversion algorithms, markedly improves the resolution and accuracy of subsurface models. Standard inversion methods typically process limited data sets, resulting in lower resolution and ambiguous interpretations. In contrast, Ultra-High Definition (UHD) inversion techniques leverage comprehensive datasets from deep and ultra-deep resistivity measurements, delivering significantly enhanced imaging capabilities. UHD inversion provides detailed and precise subsurface models, clearly delineating formation boundaries, fluid saturations, and structural complexities with unprecedented clarity.

This paper explores the combined application of deep and ultra-deep resistivity technologies and their impact on the resolution of inversion models. By demonstrating their integrated use through a practical case study in Libya, we highlight how these advanced logging techniques and inversion methods enhance the accuracy and reliability of geological interpretations, ultimately contributing to better decision-making and improved operational outcomes. Detailed comparative analyses between standard inversion and UHD inversion results underscore the substantial improvements achieved in subsurface imaging quality, geological continuity mapping, and reservoir characterization.

In subsequent sections, this paper discusses the theoretical basis, operational principles, and specific technical capabilities of deep and ultra-deep resistivity technologies. Additionally, explanations of inversion methodologies are provided. The practical implications and operational benefits of these advanced technologies are illustrated using real-world data from a challenging geological environment in Libya. This comprehensive analysis serves to underscore the transformative potential of integrating deep resistivity and ultra-deep resistivity measurements in enhancing subsurface resolution, thus significantly advancing current best practices in formation evaluation and reservoir management.

Deep Azimuthal Resistivity Tool Overview

The Deep Azimuthal Resistivity tool is an advanced logging-while-drilling (LWD) system engineered to enhance reservoir navigation and wellbore placement accuracy in complex subsurface environments. Unlike conventional resistivity tools that provide limited radial depth measurements, DAR tools can detect formation boundaries up to 30 feet (9.1 meters) away from the borehole. This extended depth of investigation enables proactive geosteering decisions, helping to avoid undesired layers and optimize wellbore trajectory (Vianna, 2025).

The core of the DAR system lies in its multi-frequency, multi-component antenna array that is strategically placed close to the drill bit. This near-bit sensor positioning allows for earlier detection of approaching boundaries and improves directional sensitivity. The tool provides real-time directional resistivity data, including azimuthal gamma ray, apparent dip, and distance-to-boundary calculations. These data streams are critical for maintaining optimal wellbore position and minimizing well trajectory deviation.

In addition to azimuthal resistivity, the tool integrates gamma ray, pressure, and vibration sensors in a single sub, streamlining the acquisition of formation and drilling dynamics data. The directional resistivity measurements are particularly valuable in horizontal and high-angle wells, where proactive adjustments can significantly influence reservoir exposure and well productivity. Real-time updates from DAR tools allow geosteering teams to interpret structural dip and bedding orientations, thus avoiding non-productive intervals or structural exits.

Figure 1 illustrates a generic configuration of a deep resistivity tool with directional antennas and near-bit placement of sensors. The system architecture supports both mud-pulse and wired pipe telemetry, enabling reliable data transmission even in high-angle and long horizontal sections.

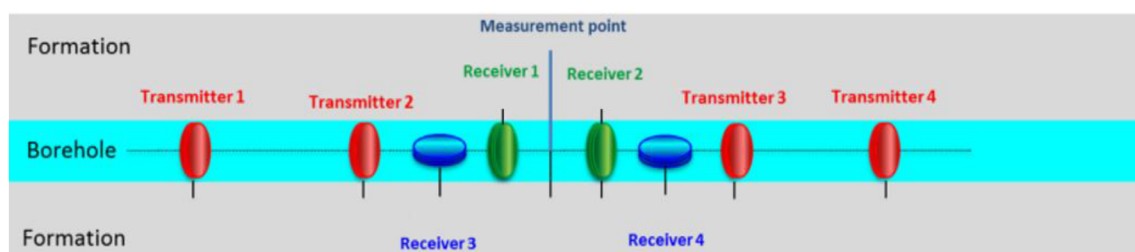


Figure 1—Generic schematic of a Deep Azimuthal Resistivity tool configuration with directional antenna array and integrated sensors.

Operationally, DAR tools function by transmitting electromagnetic (EM) signals into the formation. The interaction of these signals with the surrounding geological layers produces measurable changes in signal attenuation and phase shift. These measurements are processed through resistivity inversion models to determine the location and orientation of geological boundaries.

The DAR tool's ability to resolve resistive contrasts azimuthally and provide continuous mapping around the borehole circumference makes it exceptionally suited for thin bed evaluation and boundary tracking. According to (Zeghlache et al. 2019), the DAR tool, when combined with inversion algorithms can accurately map both the top and bottom boundaries of thin reservoir layers—even under complex resistivity profiles such as conductive zones beneath resistive anhydrite layers.

Field experience documented in (Vianna, 2025) demonstrates the tool's effectiveness in diverse lithologies, enabling high-resolution geosteering and detection of subtle formation transitions. The ability to characterize formation dips, heterogeneities, and remote boundaries has been instrumental in increasing reservoir contact and improving net-to-gross ratios, especially in environments where conventional resistivity tools fall short.

In summary, DAR technology marks a step-change in subsurface boundary mapping by offering:

- Extended depth of detection (~30 ft)
- Real-time distance-to-boundary and azimuthal sensitivity
- Integrated near-bit measurements for proactive trajectory correction
- Compatibility with wired pipe telemetry for high-resolution data acquisition

These capabilities are foundational for the successful implementation of high-definition inversion workflows and enable the accurate placement of wellbores in challenging geological settings.

Ultra-Deep Azimuthal Resistivity (UDAR) Tool Overview

The Ultra-Deep Azimuthal Resistivity (UDAR) tool represents an evolution in LWD technologies, designed explicitly to significantly extend resistivity measurements far beyond conventional limitations. UDAR provides a depth of investigation reaching up to 200 feet (60.9 meters), allowing early detection and precise delineation of geological boundaries and fluid contacts far ahead of the drill bit.

Central to UDAR's performance is its complex antenna array, comprising multiple transmitters and receivers strategically arranged coaxially and orthogonally. These antennas emit low-frequency electromagnetic (EM) signals, generating detailed three-dimensional resistivity images. This multi-component approach allows for superior azimuthal sensitivity, significantly improving the detection and characterization of formation resistivity contrasts, fluid interfaces, and structural anomalies (Potshangbam et al., 2024).

Operating typically at dual frequencies of 20 kHz and 50 kHz the tool enables multi-layered detection of geological boundaries, providing simultaneous resistivity mapping of several layers at different radial distances. This dual-frequency approach ensures robust detection of both proximal and distant geological features, facilitating a comprehensive understanding of subsurface conditions during drilling operations (Seddik et al., 2024).

A distinctive advantage of the UDAR tool lies in its capability to simultaneously resolve multiple subsurface boundaries such as reservoir tops, bases, fluid contacts, and geological structures including faults and fractures. By offering real-time azimuthal resistivity data, UDAR significantly enhances geosteering accuracy and enables proactive drilling trajectory adjustments. This immediate feedback helps mitigate operational risks, reduce non-productive time, and ensure optimal reservoir navigation and exposure (Al-Ansari et al., 2024).

Figure 2 provides a detailed schematic representation of an UDAR bottom hole assembly (BHA) configuration, highlighting the arrangement and functions of transmitter and receiver antennas, as well as integrated formation evaluation sensors.

From an operational perspective, the real-time data provided by the UDAR tool offers critical early warning of approaching subsurface hazards, geological complexities, and reservoir boundaries. This capability significantly aids in strategic and timely decision-making, particularly in horizontal and high-angle wells, where precision drilling and accurate reservoir navigation are critical for successful operational and economic outcomes (Al-Ansari et al., 2024).

In summary, the Ultra-Deep Azimuthal Resistivity technology sets a new standard in resistivity-based formation evaluation by combining extraordinary depth of investigation, high-resolution azimuthal imaging, and advanced multi-boundary inversion capabilities. These features collectively enhance reservoir navigation, drilling safety, and operational efficiency in complex geological environments.

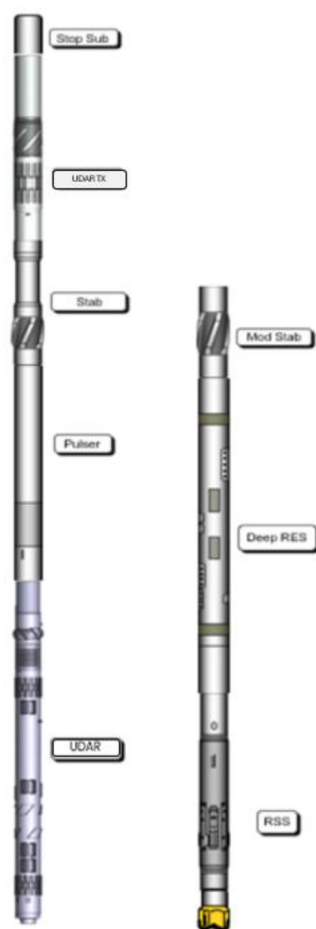


Figure 2—Detailed schematic illustration of the Ultra -Deep Azimuthal Resistivity (UDAR) tool, showing antenna arrays configured for extended-depth resistivity measurements and integrated sensors for enhanced geological and drilling evaluation.

Inversion Methodology and UHD Inversion Comparison

In resistivity logging, inversion methodologies play a pivotal role in accurately interpreting subsurface data. Traditional inversion techniques such as the Multi-Component While Drilling (MCWD) approach typically operate under simplified one-dimensional (1D) assumptions of layer continuity and isotropy. Although computationally efficient, these methods can face significant limitations in complex geological scenarios characterized by anisotropy, heterogeneity, and sharp lateral resistivity variations (Zeghlache et al., 2019; Seydoux et al., 2014).

To address these challenges, advanced Ultra-High Definition (UHD) inversion techniques have been developed, notably enhancing the resolution and accuracy of resistivity imaging. The UHD inversion employs the robust MCWD inversion software, featuring a new dynamic regularization step that significantly improves feature resolution and lateral continuity. This approach leverages high-density data from deep and ultra-deep resistivity tools, incorporating multi-component, multi-frequency electromagnetic data for superior spatial and azimuthal resolution (Vianna, 2025; Al-Ansari et al., 2024).

A crucial advancement within the UHD deterministic inversion workflow is the application of weighted resistivity data. Raw resistivity data inherently exhibit varying sensitivity and resolution across depth due to instrument limitations, geological complexity, and measurement physics. By applying specific weighting strategies, the inversion process effectively balances contributions from resistivity measurements with different depths of detection, sensitivities, and data qualities. Depth-based weighting enhances sensitivity to deeper subsurface features, quality-based weighting prioritizes reliable data by assigning higher weights to

lower-noise measurements, and resolution-balanced weighting ensures a consistent resolution model across various depths.

Despite these advancements, interpreting resistivity data still faces significant challenges, such as depth sensitivity reduction, resolution trade-offs, and noise influence. To manage these complexities, a second regularization step using a dynamic sliding window approach is employed. This method systematically scans the inverted resistivity model, calculates local statistics such as mean and standard deviation, and dynamically adjusts window sizes based on local geological variations. This approach effectively identifies and mitigates outliers, resulting in cleaner, more geologically consistent models, enhancing the stability and The practical benefits of UHD inversion have been validated in multiple field applications, demonstrating improved detection of thin beds, subtle geological features, and detailed structural mapping compared to conventional inversion methods. This enhanced resolution is especially beneficial for horizontal and high-angle drilling operations, significantly optimizing reservoir navigation, reducing drilling risks, and enhancing overall hydrocarbon recovery potential (Vianna, 2025).

Integration of DAR and UDAR for Enhanced Inversion Resolution

In complex reservoir environments, achieving high-resolution and accurate inversion models is essential for precise well placement and reservoir management. While Ultra-Deep Azimuthal Resistivity provides extended depth of investigation, integrating it with Deep Azimuthal Resistivity measurements enhances the resolution, continuity, and reliability of inversion results.

The primary advantage of combining DAR and UDAR lies in their complementary depth coverage and data density. UDAR captures broader, deeper formation features up to 200 feet, whereas DAR offers finer, higher-resolution azimuthal data closer to the borehole (up to 30 feet). When integrated within a hybrid inversion framework, these datasets significantly improve the fidelity of resistivity models, allowing better delineation of thin layers, sharper boundary definition, and more accurate thickness estimation.

To operationalize this integration, a customized inversion workflow enables field engineers to select and merge resistivity curves from both the UDAR and DAR tools—ranging from ultra-deep frequencies (20 kHz) up to high-frequency bands (2 MHz). Through this interactive interface, individual frequency components or azimuthal channels (e.g., ZX components) can be assigned distinct weights or noise tolerances. For instance, increasing the weight (or lowering the modeled noise) on higher-frequency azimuthal curves emphasizes near-wellbore resolution, whereas emphasizing low-frequency UDAR channels extends the depth of detection. By dynamically adjusting relative noise and weight parameters in the inversion configuration, the algorithm tailors the influence of each dataset to match specific well objectives—whether prioritizing sharper bed resolution or deeper boundary detection. This workflow flexibility is the key innovation of the approach, allowing real-time tuning of the hybrid inversion to maximize geological insight and geosteering performance.

Figure 3 illustrates this improvement vividly through a side-by-side comparison:

- The top panel shows the inversion result using UDAR data alone, displaying lower resolution and some discontinuity in layer boundaries.
- The bottom panel shows the inversion with combined DAR and UDAR inputs, revealing enhanced lateral continuity, finer boundary clarity, and more precise layer thicknesses.

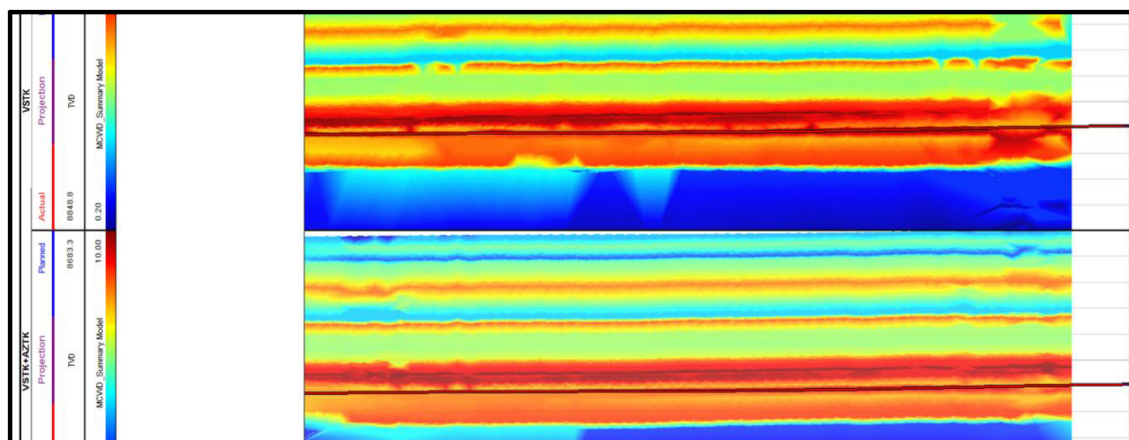


Figure 3—Comparative inversion results demonstrating improved resolution, continuity, and accuracy achieved by integrating DAR and UDAR data

This integration enables geosteering teams to confidently interpret complex geological features, such as thin laminations and subtle resistivity contrasts, which are critical in reservoirs with intricate stratigraphy or fluid distributions.

Moreover, the increased data density and resolution achieved through this combination reduce geological uncertainties and risk, thereby minimizing non-productive time and improving hydrocarbon recovery potential.

Such combined inversion techniques have been validated in multiple field applications, including the Libya Sarir field, demonstrating the practical benefits of this approach for challenging reservoir navigation and optimized well placement (Al-Ansari et al., 2024).

Case Study

This case study involves operations conducted in Libya's Sarir Oilfield, characterized by heterogeneous carbonate reservoirs with complex stratigraphy, significant lateral variability in reservoir quality, and challenging structural conditions, including faults and fractures. These geological challenges necessitated advanced logging technologies for optimal well placement and precise reservoir navigation.

The operational objective was to accurately land and geosteering a horizontal well within a thin, complex carbonate reservoir approximately 150 feet thick. Critical to success was maintaining precise well trajectory control to continuously position the wellbore within the highest-quality productive reservoir interval while avoiding adjacent water-bearing zones.

To achieve these objectives, an integrated approach combining UDAR and DAR tools was employed. The UDAR tool, with its extended depth of investigation (up to 200 feet), provided early and precise detection of geological boundaries, including reservoir tops, bases, and fluid contacts, allowing proactive trajectory adjustments far ahead of the drill bit. Concurrently, DAR, with its superior near-wellbore resolution and azimuthal sensitivity (up to 30 feet), enabled detailed characterization of thin layers and fine-scale resistivity contrasts close to the borehole.

Integrating DAR and UDAR data into a unified hybrid inversion framework significantly enhanced the resolution and accuracy of the inversion model. The combined inversion clearly delineated thin beds, subtle geological features, and reservoir boundaries, providing a high-definition subsurface image essential for effective geosteering decisions.

The successful deployment of the integrated DAR-UDAR technology delivered exceptional results. Real-time inversion data from this combined approach enabled precise and proactive geosteering, ensuring continuous wellbore positioning within the target reservoir zone throughout drilling operations. This

A visual representation of this achievement is illustrated in [Figure 4](#), showing the curtain section of the drilled well trajectory with consistently accurate reservoir contact facilitated by the combined DAR-UDAR inversion. Additionally, [Figure 5](#) depicts a detailed 3D visualization of the UHD inversion results, clearly highlighting resistivity variations and structural details accurately captured through this integrated methodology.



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Conclusions

The study demonstrates the significant benefits of integrating Deep Azimuthal Resistivity (DAR) with Ultra-Deep Azimuthal Resistivity (UDAR) technologies for enhancing the resolution and accuracy of inversion results in complex reservoir conditions. The detailed case study from Libya's Sarir Oilfield illustrates the practical advantages of this integrated approach, achieving precise geosteering, maximum reservoir contact, and improved operational efficiency. Combining DAR and UDAR data in a hybrid inversion framework substantially enhances geological interpretation, reduces uncertainties, and optimizes drilling strategies. Thus, the integration of these advanced resistivity technologies represents a best-practice solution for accurate well placement and effective reservoir management in geologically challenging environments.

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